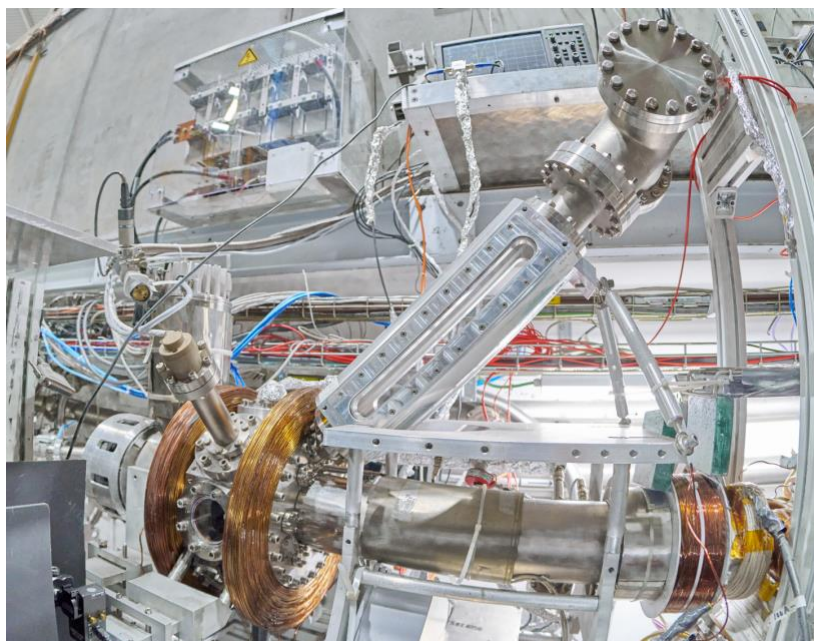


AEgIS experiment paves the way for new set of antimatter studies by laser-cooling positronium

In cooling positronium with laser light for the first time, AEgIS may also have taken the first step towards a matter–antimatter system that emits laser-like gamma-ray light



The set-up used by the AEgIS team to laser-cool positronium. (Image: CERN)

[AEgIS](#) is one of several experiments at CERN's Antimatter Factory producing and studying antihydrogen atoms with the goal of testing with high precision whether [antimatter](#) and matter fall to Earth in the same way. In a [paper](#) published today in Physical Review Letters, the AEgIS collaboration reports an experimental feat that will not only help it achieve this goal but also pave the way for a whole new set of antimatter studies, including the prospect to produce a gamma-ray laser that would allow researchers to look inside the atomic nucleus and have applications beyond physics.

To create antihydrogen (a positron orbiting an antiproton), AEgIS directs a beam of positronium (an electron orbiting a positron) into a cloud of antiprotons produced and slowed down in the Antimatter Factory. When an antiproton and a positronium meet in the antiproton cloud, the positronium gives up its positron to the antiproton, forming antihydrogen. Producing antihydrogen in this way means that AEgIS can also study positronium, an antimatter system in its own right that is being investigated by experiments worldwide.

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Maintaining motivation throughout your career

Positronium has a very short lifetime, annihilating into gamma rays in 142 billionths of a second. However, because it comprises just two point-like particles, the electron and its antimatter counterpart, “it’s a perfect system to do experiments with”, says AEgIS spokesperson Ruggero Caravita, “provided that, among other experimental challenges, a sample of positronium can be cooled enough to measure it with high precision”.

This is the feat accomplished by the AEgIS team. By applying the technique of laser cooling to a sample of positronium, the collaboration has already managed to more than halve the temperature of the sample, from 380 to 170 degrees kelvin. In follow-up experiments the team aims to break the barrier of 10 degrees kelvin.

AEgIS’ laser cooling of positronium opens up new possibilities for antimatter research. These include high-precision measurements of the properties and gravitational behaviour of this exotic but simple matter–antimatter system, which could reveal new physics. It also allows the production of a positronium Bose–Einstein condensate, in which all constituents occupy the same quantum state. Such a condensate has been proposed as a candidate to produce coherent gamma-ray light via the matter-antimatter annihilation of its constituents – laser-like light made up of monochromatic waves that have a constant phase difference between them.

“A Bose-Einstein condensate of antimatter would be an incredible tool for both fundamental and applied research, especially if it allowed the production of coherent gamma-ray light with which researchers could peer into the atomic nucleus.” says Caravita.

Laser cooling, [which was applied to antimatter atoms for the first time](#) about three years ago, works by slowing down atoms bit by bit with laser photons over the course of many cycles of photon absorption and emission. This is normally done using a narrowband laser, which emits light with a small frequency range. By contrast, the AEgIS team uses a broadband laser in their study.

“A broadband laser cools not just a small but a large fraction of the positronium sample,” explains Caravita. “What’s more, we carried out the experiment without applying any external electric or magnetic field, simplifying the experimental set-up and extending the positronium lifetime.”

The AEgIS collaboration shares its achievement of positronium laser cooling with an independent team, which used a different technique and posted their result on the [arXiv](#) preprint server on the same day as AEgIS.

About AEgIS:

The AEgIS collaboration is composed of several research groups from CERN, Istituto Nazionale di Fisica Nucleare (units of Milano, Pavia and the Trento Institute for Fundamental Physics and Applications), the University of Oslo, the Universite Paris-Saclay and the Centre National de la Recherche Scientifique, the University of Liverpool, the Warsaw University of Technology, the University of Trento, the Jagiellonian University of Krakow, the Raman Research Institute of Bangalore, the University of Innsbruck, the University and the Politecnico of Milan, the University of Brescia, the Nicolaus Copernicus University in Torun, the University of Latvia, the Institute of Physics of the Polish Academy of Sciences and the Czech Technical University of Prague.

Accelerator Report: Advancing smoothly through recommissioning and overcoming challenges

On 16 February, the symbolic LHC key was officially handed over to the Operations team, marking the start of the hardware recommissioning phase for the LHC. Early this week, 8600 hardware tests were successfully

executed, out of the 10 336 tests that have to be performed. The injection of the first protons into the LHC is scheduled for 11 March, but could be brought forward by a few days if all goes well.

Meanwhile, the commissioning of the LHC injector chain is advancing smoothly. Linac4 is consistently delivering beams to the PS Booster, which has now set up and tuned all operational beams for downstream users. The PS has also set up the initial beams required by the SPS, including a single-bunch LHC beam and a low-intensity version of the beam that is destined for the SPS North Area experiments.

On the SPS front, hardware commissioning is proceeding according to plan, though not without some challenges. The “heat run”, a critical step in the commissioning process, involves pulsing the main magnets intensively for about 12 hours, with a current close to the maximum average current the magnets can operate with. The aim of the heat run is to detect any anomalies in the cooling of the magnets. Under normal conditions, the magnets will reach their steady state temperature after 20 to 30 minutes.

Last week, only 10 minutes after the beginning of the heat run, the operators in the CERN Control Centre (CCC) received an alarm: one or more magnets in one of the SPS sextants (one sixth of the machine’s circumference) had overheated. The magnets are protected against overheating; this protection cuts the electrical current circulating through the magnets to avoid any damage. A prompt intervention by experts equipped with thermal cameras identified the problematic magnet, and an investigation revealed that rubber debris from an anti-return valve was obstructing the cooling water flow. The issue was addressed by cleaning all 110 filters in the affected sextant and flushing the circuit, ensuring that no rubber debris was left. A second

heat run confirmed that all the magnets are now operating at normal temperatures.



This thermal image clearly shows a significant temperature difference between the magnet on the left (in blue) and the overheating magnet on the right (in yellow). (Image: CERN)



The water filter blocked with rubber debris. (Image: CERN)

Overall, both hardware and beam commissioning activities across the accelerator complex are progressing well. Despite some issues, typical of the annual recommissioning, nothing has jeopardised the beam delivery schedule so far.

Rende Steerenberg

CERN70: The heart of CERN’s accelerator chain

Part 4 of the [CERN70](#) feature series. Günther Plass, former Director of Accelerators at CERN, joined the Magnets group at the Proton Synchrotron (PS) in 1956. Three years later, the machine went into service and became the most powerful accelerator in the world

In 1957, CERN staff moved into the new buildings on the Meyrin site in Geneva, and the rooms were rapidly filled with equipment for the Proton Synchrotron (PS).

By the end of July 1959, assembly of the PS – the accelerator is more than 600 metres in circumference! – was completed and, on 16 September, the first beam was circulated. On 24 November 1959, the PS accelerated protons for the first time to its nominal energy of 24 gigaelectronvolts (GeV).



Aerial view of the CERN site and the PS ring in April 1959 (Image: CERN)

The machine is based on a revolutionary new concept developed at Brookhaven National Laboratory in the United States, which makes it possible, for the same budget, to achieve a much higher particle energy than with a traditional synchrotron. In this way, the young Laboratory demonstrated its ability to translate a new concept into reality.

Even today, 65 years later, the PS is the heart of CERN's accelerator chain. And while the machine has of course undergone many phases of improvement, its fundamental structure has remained unchanged.

Recollections

Günther Plass joined the Proton Synchrotron (PS) team in 1956 and worked for 25 years on PS-related assignments. In the 1980s, he became Deputy Project Leader of the Large Electron Positron Collider (LEP), before becoming CERN's Director of Accelerators.

"A dream began for me in autumn 1954 at the annual meeting of the German Physical Society. At that gathering, an enthusiastic Werner Heisenberg spoke of recent meetings preparing for CERN, a European Laboratory for Fundamental Physics just being launched in Geneva. A pioneering European enterprise, the new laboratory was to build a particle accelerator hundreds of metres in circumference, huge compared to the cyclotrons I

had been reading about at the time. Physicists from a dozen European countries would work together on the construction, and later the use, of this almost unbelievable machine, which would stretch technologies to their very limits.

I certainly wasn't the only one to be deeply impressed by this report. I even nurtured the hope that my moderate command of languages might one day help me to participate in that enterprise. Back in 1950, I had spent several weeks on a language course in Dijon, France, and in 1951 I spent another few weeks picking potatoes along with students from several countries near Hull in the United Kingdom.



Completion of the first PS magnet unit was celebrated by photographing the entire Magnet group sitting on it. This first magnet unit was named Margherita, after group member Margherita Cavallaro, shown here holding a sign. Günther Plass is fourth from the left. (Image: CERN)

About one and a half years after Heisenberg's presentation, I learned that my current [magnet] specialisation was in demand for CERN's project, so I rushed off my application and, much to my surprise, was offered a job. On 18 June 1956, I dutifully reported to the PS Division's secretariat at the Institute of Physics. I was directed to join my future colleagues in the lecture hall, where a Symposium on Particle Accelerators and Pion Physics was under way. I thus began my career by listening to talks about pions, muons and strange particles — what a mysterious new world I had come into!"



On 25 November 1959, in front of the CERN staff gathered in the Main Auditorium, John Adams, then Director of the PS division, held a bottle of vodka in his hands. The bottle had been given to him a few months earlier on a trip to Dubna, in the Soviet Union, where the world's most powerful accelerator, the Synchrophasotron, had just been commissioned. The bottle was only to be opened if Dubna's record energy of 10 GeV was exceeded. On 24 November, the PS circulated protons at more than twice that energy, 24 GeV. Before sending the empty bottle back to the Soviet Union, John Adams placed inside it a photograph of the 24 GeV pulse, as proof of the record. (Image: CERN)

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary. Günther Plass [passed away in 2020](#), aged 90.

Work well feel well: Suffer or enjoy?

Part 3 of the Work Well Feel Well series looks at recognising and learning from emotion



Sphere of control

What we experience in life can be out of our control. What we can control, however, is **how we react**. This doesn't mean staying positive in every situation and hence suppressing or ignoring how we really feel. **All emotions are valid** and deserve to be acknowledged and accepted. They provide **useful signposts**, such as anger indicating when a boundary has been crossed.

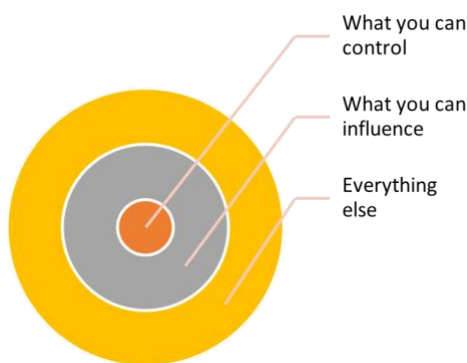
In challenging situations, it helps to understand what we can control or influence, and what we

can't. This **sphere of control** gives us the clarity to move forward, **changing what we can** and **accepting what we can't**.

We can take back more control of a situation by reaffirming our boundaries and saying no. There are many examples of **saying no respectfully** to protect ourselves from becoming overloaded. Here are just a few:

- Thank you for thinking of me, but I can't.
- Unfortunately, it's not a good time.
- It isn't possible with my current workload.
- I have other priorities that are more important at the moment.
- I'm not able to prioritise that right now.
- I already have plans. Perhaps next time.

We experience a range of emotions each day. Taking a moment to **observe our emotions** and **develop a plan or intention** (see exercise below) that is respectful of our needs, can guide and help us to enjoy a balanced position to external pressures.



Take action

As part of the “Efficiency and caring at work” campaign, the [Work Well Feel Well website](#) now offers useful resources that can be downloaded, including an [exercise](#) on observing emotions, defining needs and developing a plan or intention.

In addition:

- The [recording](#) of the recent Micro-talk #8 “Ego vs Innovation : le regard des neurosciences cognitives” (in French) reinforces how we can learn from our emotions and our errors.
- The CERN Medical Service [mental health support](#) resources include a self-assessment and contact details of the CERN psychologists.
- The [CERN Ombud](#) articles “[feeling beyond help](#)” and “[put your own oxygen mask on first](#)” are also relevant to this topic.

HR department

This is the third of a 12-part [Work Well Feel Well](#) series, with articles publishing every two months.

A new data centre at CERN

The new data centre complements the existing centre and will allow CERN to respond to the growing data-processing needs of the worldwide scientific community

On 23 February 2024, a brand-new data centre was inaugurated on CERN's Prévessin site (France), marking the completion of a major project for the Organization's computing strategy. Spanning more than 6000 square metres and including six rooms for IT equipment with a cooling capacity of 2 MW each, the centre will host CPU (central processing unit) servers for physics data processing as well as a small amount of CPU servers and storage capacity for business continuity and disaster recovery (for example,

when data is corrupted). CERN's main data centre on the Meyrin site (Switzerland) will continue to house the majority of the Organization's data storage capacity.

The rate of data production of the experiments at the Large Hadron Collider (LHC) continues to grow, already reaching some 45 petabytes per week, and this is expected to double in the era of the High-Luminosity LHC, the major upgrade of CERN's current flagship accelerator, the LHC. The data from these experiments is fed into the Worldwide

LHC Computing Grid (WLCG), a collaboration of around 170 data centres distributed across more than 40 countries, with a storage capacity of about 3 exabytes and one million CPU cores distributed across the network. While the Meyrin data centre has so far performed the Tier 0 role, that is, the core for the LHC Computing Grid, the Prévessin centre will provide vital additional computing capacity to CERN.



The inauguration of the new data centre in Prévessin. From left to right: Pippa Wells, CERN's Deputy Director for Research and Computing; Charlotte Warakaulle, CERN's Director for International Relations; Aurélie Charillon, Mayor of Prévessin-Moëns; Joachim Mnich, CERN's Director for Research and Computing; Yves Nussbaum, Director Marché Industrie, AXIMA; and Enrica Porcari, Head of Information Technology Department at CERN. (Image: CERN)

The new building was built in a record time of less than two years. It complies with strict technical requirements to ensure its environmental sustainability, and is equipped with an efficient heat-recovery system that will contribute to heating buildings on the Prévessin site.

The backbones of our interconnected world, data centres are energy-intensive infrastructures. According to a recent report, their energy consumption accounts for about 1.5% of the European Union's total electricity consumption. Two parameters characterise the environmental sustainability of a data centre: the power usage effectiveness (PUE) – the ratio of total data centre input power to IT load power – and the water usage effectiveness (WUE) – the ratio between the use of water in data centre systems and the energy consumption of the IT equipment.

The new Prévessin centre has a PUE target of 1.1, lower than the worldwide average of 1.6, and close to 1.0, which would be the value for a perfectly efficient data centre, where all the power is delivered to the IT equipment.

It has a WUE target of 0.379 litres per kWh thanks to an innovative water recycling system. The cooling system will be automatically triggered when the outside temperature reaches 20 degrees Celsius. Five huge fan-walls installed in each room will ensure that the overall temperature does not exceed 32 degrees Celsius.

The new centre was designed, built and will be operated in the framework of a FIDIC (International Federation of Consulting Engineers) Gold Book contract, which also ensures its financial sustainability. The building's IT rooms will gradually be equipped with up to 78 racks each. Starting from the top-floor rooms, they are expected to be fully equipped over the next ten years.

Antonella Del Rosso

ABB and CERN identify 17.4% energy-saving opportunity in the Laboratory's cooling and ventilation motors

Through a strategic research partnership focused on CERN's cooling and ventilation systems, energy efficiency audits have helped to identify a savings potential of 17.4% across a total of 800 motors

In a joint research project conducted between 2022 and 2023, [ABB](#), a global company specialised in digitally enabled motor and drive solutions to support a low-carbon future for industry, infrastructure and transportation, and CERN

developed a roadmap for reducing the energy consumption of CERN's cooling and ventilation systems via data-driven energy efficiency audits. These systems are responsible for the cooling and ventilation of CERN's accelerator complex,

experimental areas and data centres. The roadmap identified potential annual energy savings of up to 31 gigawatt-hours (GWh). If achieved, these savings could be enough to power more than 18,000 European households⁽¹⁾ and could avoid 4 kilotonnes of CO₂ emissions⁽²⁾, the same as planting 420,000 trees⁽³⁾.

Energy efficiency audits involve evaluating the performance and efficiency of motors, based on their operating data. Such audits help large facilities like CERN to identify the most significant energy-saving opportunities across whole groups of motors. CERN and ABB experts assessed a wide variety of data from motors used for various cooling and ventilation applications. They combined data from multiple sources, including digitally connected motors, CERN's supervisory control and data acquisition (SCADA) system, which is responsible for the control and monitoring of the cooling and ventilation installations, and data gathered directly from pumps, piping and instrumentation. The experts analysed the efficiency of the whole system in order to pinpoint the motors that present the best business case for energy efficiency upgrades.

Giovanni Anelli, Head of [CERN's Knowledge Transfer group](#), said, "The collaboration with ABB was set up with the aim of optimising the Laboratory's cooling and ventilation infrastructure to reduce its energy consumption, and is in line with CERN's commitment to minimise its

environmental footprint as well as to share the findings publicly for the benefit of society. It's an excellent example of collaboration where each side brings its own contribution to the table. CERN brings its large-scale infrastructure and ABB contributes with its technology and service expertise. We are very happy with the final result of this research project as we have exceeded our goal of identifying a 10-15% energy efficiency improvement."

"We are proud to cooperate with CERN and to support its goal to conduct physics research with a low-carbon footprint by helping it to improve the energy efficiency of its cooling and ventilation systems," said Erich Labuda, President of the Motion Services division at ABB.

CERN's next step will be to selectively upgrade motors with the highest energy-saving potential, based on the data collected during the audit.

Read the press release published by ABB here:
<https://cern.ch/itav5>

CERN Knowledge Transfer group

(1) [EU average](#) (~1670 kWh/year)

(2) [Electricity Maps](#) | [Live 24/7 CO₂ emissions of electricity consumption](#)

(3) [How Much CO₂ Does A Tree Absorb? – One Tree Planted](#)

CMS collaboration explores how AI can be used to search for partner particles to the Higgs boson

This search uses computer vision techniques to look for collimated bursts of light, a potential signature of Higgs partner particles

As part of their quest to understand the building blocks of matter, physicists search for evidence of new particles that could confirm the existence of physics beyond the Standard Model (SM). Many of these beyond-SM theories postulate the need for additional partner particles to the Higgs boson. These partners would behave similarly to the SM Higgs boson, for example in terms of their "spin", but would have a different mass.

To search for Higgs partner particles, scientists at the CMS collaboration look for the signatures of these particles in the data collected by the detector. One such signature is when the particles decay from a heavy Higgs partner (X) particle to two lighter partner particles (ϕ), which in turn each decay into collimated pairs of photons. Photon signatures are ideal to search for particles with unknown masses as they provide a clean,

well-understood signature. However, if the ϕ is very light, the two photons will significantly overlap with each other and the tools usually applied for the photon identification fall apart.

This is where artificial intelligence (AI) comes in. It is well known that machine learning computer vision techniques can differentiate between many faces, and now such AI methodologies are becoming useful tools in particle physics.

The CMS experiment searched for the X and ϕ partners of the Higgs boson using the hypothetical process $X \rightarrow \phi\phi$, with both ϕ decaying to collimated photon pairs. To do this, they trained two AI algorithms to distinguish the overlapping pairs of photons from noise, as well as to precisely determine the mass of the particle from which they originated. A wide range of masses was explored. No evidence for such new particles was seen, enabling them to set upper limits on the production rate of this process. The result is the most sensitive search yet performed for such Higgs-like particles in this final state.

How can the scientists test the AI's effectiveness? It is not as easy as verifying AI facial differentiation, where you can simply check by looking. Thankfully, the SM has well-understood processes, which CMS physicists used to validate and control the AI techniques. For example, the η meson, which also decays to two photons, provided an ideal test bench. Scientists at CMS were able to cleanly identify and reconstruct the η meson when searching for its decay into photons when they applied these AI techniques.

This analysis clearly shows that AI algorithms can be used to cleanly identify two-photon signatures from the noise and to search for new massive particles. These machine learning techniques are continuously improving and will continue to be used in unique analyses of LHC data, extending CMS searches to even more challenging cases.

CMS collaboration

Read more here: <https://cern.ch/2w2ay>

Hearing the sound of quark–gluon plasma

The CMS collaboration has measured the speed of sound in the quark–gluon plasma more precisely than ever before, offering new insights into this extremely hot state of matter

Neutron stars in the Universe, ultracold atomic gases in the laboratory, and the quark–gluon plasma created in collisions of atomic nuclei at the Large Hadron Collider (LHC): they may seem totally unrelated but, surprisingly enough, they have something in common. They are all a fluid-like state of matter made up of strongly interacting particles. Insights into the properties and behaviour of any of these almost perfect liquids may be key to understanding nature across scales that are orders of magnitude apart.

In a new paper, the CMS collaboration reports the most precise measurement to date of the speed at which sound travels in the quark–gluon plasma, offering new insights into this extremely hot state of matter.

Sound is a longitudinal wave that travels through a medium, producing compressions and rarefactions of matter in the same direction as its movement. The speed of sound depends on the

medium's properties, such as its density and viscosity. It can therefore be used as a probe of the medium.

At the LHC, the quark–gluon plasma is formed in collisions between heavy ions. In these collisions, for a very small fraction of a second, an enormous amount of energy is deposited in a volume whose maximum size is that of the nucleus of an atom. Quarks and gluons emerging from the collision move freely within this area, providing a fluid-like state of matter whose collective dynamics and macroscopic properties are well described by theory. The speed of sound in this environment can be obtained from the rate at which pressure changes in response to variations in energy density or, alternatively, from the rate at which temperature changes in response to variations in entropy, which is a measure of disorder in a system.

In heavy-ion collisions, the entropy can be inferred from the number of electrically charged particles emitted from the collisions. The temperature, on the other hand, can be deduced from the average transverse momentum (i.e. the momentum transverse to the collision axis) of those particles. Using data from lead–lead collisions at an energy of 5.02 trillion electronvolts per pair of nucleons (protons or neutrons), the CMS collaboration has measured for the first time how the temperature varies with the entropy in central heavy-ion collisions, in which the ions collide head on and overlap almost completely.

From this measurement, they obtained a value for the speed of sound in this medium that is nearly

half the speed of light and has a record precision: in units of the speed of light, the squared speed of sound is 0.241, with a statistical uncertainty of 0.002 and a systematic uncertainty of 0.016. Using the mean transverse momentum, they also determined the effective temperature of the quark–gluon plasma to be 219 million electronvolts (MeV), with a systematic uncertainty of 8 MeV.

The results match the theoretical expectation and confirm that the quark–gluon plasma acts as a fluid made of particles that carry enormous amounts of energy.

CMS collaboration

Read more here: <https://cern.ch/q3q67>

We love your CERN poems

CERN community, we asked you to send us your CERN-related Valentine’s poems, thank you for all the entries that we received! It wasn’t easy to pick a favourite, but in the end, here is our winning English poem:

Ninety five percent of the **Universe**
has not been discovered, yet
My search for perfection stopped
when I saw you – si parfaite

You **Opened** the **Gateway** to my heart
You’re in all my **Ideas**, fair and **Square**
The force that draws me to you
Is stronger than the ions from **LEIR**

I had my whole week blocked
To just look at you in awe
until you broke my heart by saying
“I work in Preveessin”

When it comes to you,
I don’t have to think twice
Will you meet me halfway
At **CERN Animal Shelter for Computer Mice?**

In the end it’s you and me,
Surrounded by all this neige
Our own little Quantum Entanglement
Where **Data becomes Knowledge**

By Varsha Bhat (IT/CD)

A special mention also goes to the CMS and ATLAS haikus:

What is CMS?
Compact Muon Solenoid.
The best detector.

By Sofia Hurst (EP/CMX)

I love you, ATLAS.
CMS got long bus rides.
We got point 1. ha.

By Rea Thornberry (EP/UAT)

The authors of the winning English poem and winning French poem each receive a goodie bag from the CERN shop including “I love CERN” socks, an attractive magnetic pencil and a Standard Model notebook to compose more odes to technology.

Thank you once again to everyone who entered. The full list of poems is available here: <https://cern.ch/2024-poems> (CERN login required).

Internal Communication

Indico: 20 years of event managing

The CERN-developed open source software, Indico, is being chosen by a growing number of users worldwide. As it celebrates 20 years, find out what new features are planned for 2024

Thanks to 300 active servers located in 52 different countries, some 400 000 users can experience the powerful and user-friendly features that Indico offers to any event organiser. Born about twenty years ago with the “[Event 0](#)” – the CHEP 04 scientific conference – today Indico serves not only the large community of particle physicists working at CERN but also a plethora of other institutions, including the JACoW.org conferences and the UN.

Indico started as a follow-up to the AgendaMaker tool that a small team of developers had created for the ATLAS collaboration, which today is still one of the most active users of the platform. “Following that initial request, a funding request was made to the EU to evolve from a meeting-oriented application to a catch-all platform where we could host a one-hour meeting as well as a one-week conference with parallel sessions,” recalls Jean-Yves Le Meur, in charge of the early developments of Indico. “Since the beginning, Indico was designed to be modular, and I am amazed to see today how the successive managers of this system have extended and enriched its features in all directions.”

In 2023, the Indico application managed some 145 000 events worldwide and accommodated a regular flow of requests coming from the community, whose members also contribute to the development of the open source software package. Last December, a big release – Indico 3.3 – was made available on the CERN-managed instances, and the release for the general public will follow in the coming months. “The release has brought a lot of improvements compared to previous versions,” explains Adrian Mönnich, Indico's project manager and lead developer. “For example, thanks to contributions from the UN, we have improved the accessibility of Indico. Indeed, the application is now more suitable for visually

impaired people who use screen readers. The existing [Indico check-in mobile application](#) has also been completely rewritten, and Indico now supports the generation of fully customisable PDF documents, including receipts, certificates of attendance and any other document that the user might need. Once generated, the documents can be downloaded from the registration page.”

The continuous Indico upgrade never stops, and here is a sneak peek of what we can expect in 2024: “We plan to release a brand-new conference timetable interface, which will replace the current one that dates back to 2008,” says Pedro Ferreira, leader of the Conferencing Technologies section. “The team is also working on a new integration with the next-generation conferencing rooms, and we hope to make significant steps towards a refreshed Indico user interface including greater mobile friendliness.”

Indico is a CERN-developed open source software. Since last year, a new governance model has been in place to encourage contributions from partner institutions. Find more technical information here: <https://getindico.io/>.

Events and you

Find relevant Indico events and what's on at CERN and beyond:

- [Upcoming events](#) is a curated list of events published in every *Bulletin*
- [CERN Campus app](#) now includes events
- [Home.cern ongoing and upcoming events list](#)
- Indico also shows ALL events happening [today](#), [this week](#) and [in a calendar view](#).

Antonella Del Rosso

ATLAS congratulates its 2023 Thesis Awards winners



From left to right: ATLAS collaboration Board Chair Maria Jose Costa; ATLAS Thesis Awards Committee Chair Antonella De Santo; ATLAS Thesis Award winners Nicole Hartman, Joshua Beirer, Savannah Clawson, Hassnae El Jarrari and Xiao Yang; and ATLAS Spokesperson Andreas Hoecker. Not pictured: ATLAS Thesis Award winners Samuel Van Stroud and Prajita Bhattarai. (Image: K. Anthony/CERN)

The ATLAS collaboration celebrated the achievements of its exceptional PhD students at the recent Thesis Awards ceremony. Established in 2010, the ATLAS Thesis Awards recognise the remarkable contributions made by students to the ATLAS collaboration through their doctoral theses. Students play pivotal roles in the collaboration while gaining invaluable skills crucial to their professional pursuits.

The [2023 ATLAS Thesis Awards](#) were announced on 15 February 2024 at a ceremony held at CERN's main auditorium. The award winners are: **Joshua Beirer** from CERN & Georg-August-Universität Göttingen (Germany), **Prajita Bhattarai** from Brandeis University (USA), **Savannah Clawson** from the University of Manchester (UK), **Hassnae El Jarrari** from Université Mohammed-V De Rabat (Morocco), **Nicole Hartman** from Stanford University & SLAC (USA), **Samuel Van Stroud** from University College London (UK), and **Xiao Yang** from the University of Science and Technology of China (China).

"PhD students aren't just the beating heart of the ATLAS collaboration – they're the brains behind many of our achievements," said Antonella De Santo, Chair of the Thesis Awards Committee. "PhD students make up a significant fraction of ATLAS collaboration members and contribute to a diverse range of research areas, including physics analysis, detector operations and upgrades, and software and hardware developments. The ATLAS Thesis Awards are our way of recognizing and highlighting their outstanding achievements."

Katarina Anthony

Read the full story and explore the winning theses on the [ATLAS website](#).

Computer Security: The hardship of three security paradigms

Bravo to all those who participated in the Bull**** Bingo in the last Bulletin issue, in particular to those who sent in their solution and won a delicious Hawaiian pizza topped with pineapple and a Coke. Given the many replies, it seems that our Bingo was too easy? But that's what "security" should be: easy, straightforward, simple. Paradigm #1 – the "KISS" paradigm: "Keep it simple, stupid". Unfortunately, this simplicity is spoiled time and again by the complex computing environment at CERN mixing the divergent needs of academia (research and computing sector), administration

(finance and HR sector) and industry (the accelerator sector); by CERN's legacy of using its resources for personal business like sending/receiving private emails, hosting personal webpages, or our bring-your-own-device (BYOD) policy to connect all of your own devices to CERN's campus network; by the cacophony of historically grown systems performing similar – but not identical – tasks (CDS/CERNBox/EDMS/EOS/Google Workspaces/MyFiles/OneDrive/Sharepoint, or Kubernetes vs OpenStack vs OpenShift); and the

problems coming with the cacophony of terminating old and outdated services (like the very slow and complicated AFS and DFS migration to CERNBox, killing the old SSO for the benefit of the new one, or moving Drupal-hosted websites to WordPress). So, KISS is hardly a reality at CERN. We should strive to do better. Simpler. More homogenous. More centralised. More controlled. KISS.

Unfortunately, again, and given additional constraints – lack of resources or time pressure – paradigm #2 kicks in: “cheap, convenient, secure – pick two”. That makes security a permanent uphill struggle as nobody would pick “secure” given that “cheap” and “convenient” always trump. Would you? Instead, security is given low priority, filed to the back, and applied only when time and resources allow (or the implementers are security aware). Again, we should strive to do better. Last year’s audit on cybersecurity urged higher priority and recommended that the Organization “define and implement a process to ensure security is considered in any project” and “implement a security risk management process” under the auspices of the Computer Security team (a dedicated Bulletin article on this topic will be published soon).

Following general best practices – and as re-emphasised by the aforementioned audit – the Computer Security team has always aimed to

deploy and deepen “defence-in-depth” – paradigm #3. With your help – given that “security is everyone’s responsibility” (Bingo solution C1) – “2FA is a big step forward for account protection” (A2) and we are grateful to now have more than 10 000 accounts under this protection. On another defence level, we succeed well at dismissing malicious websites, domains and IPs on the firewall level, but struggle to filter malicious emails (and promise to improve on that during 2024). Still, we are counting on you to detect those that made it through: “Only the link behind a text/QR code reveals its truth” (B3). But we also try to help you, as “CERN’s anti-malware software is free for you to download” (E4)*. Defence-in-depth. Hard to implement, but possible to have.

Hence, these paradigms “KISS – keep it simple, stupid” and “defence-in-depth” go hand in hand once we all jointly pick the right two of “cheap, convenient, secure”. Let’s overcome the hardship imposed by these three paradigms. You and us together. Together securing the Organization. Preventing any disasters.

** The fifth solution is D5 – “Encryption is easy; key management is complicated” – but that is a technical detail being taken care of inside IT.*

CERN Computer Security Team

Official news

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Summer work for children of members of the personnel

During the period from 10 June to 6 September 2024 inclusive, there will be a limited number of jobs for summer work at CERN (normally unskilled work of routine nature), which will be made

available to children of members of the personnel (i.e. anyone holding an employment or association contract with the Organization). Candidates must be aged between 18 and 24 inclusive on the first

day of the contract and must have insurance covering illness. The duration of all contracts will be 4 consecutive weeks and the allowance will be CHF 1577 for this period. Candidates should apply via HR department's electronic recruitment system, SmartRecruiters: <https://smrtr.io/j3W95>. Completed application forms must be returned by 11 March 2024 at the latest. To allow as many people as possible to benefit, each child may participate in this programme only once.

The results of the selection will be available mid of May 2024.

For further information, please contact:
Virginie.Galvin@cern.ch – Tel. 72855
(Geraldine.Ballet@cern.ch – Tel. 74151)

HR department

Conditions of access to the fenced parts of the CERN site

Operational circular No. 2 (Rev. 3) - update of Annex I to the implementation measures

Members of the personnel and other persons concerned are hereby informed that the Director-General has approved, further to consultation with the Staff Association, an update of Annex I to the document "Implementation measures, subsidiary document to Operational Circular No. 2 (Rev. 3)".

The modifications concern the opening hours of gates A and C (access points to the fenced parts of the Meyrin site).

This update of the Annex I will enter into force on 1 March 2024 and is available via the following link: <https://cds.cern.ch/record/2109697>

Information on these new access modalities is also available from the SCE website: <https://sce-dep.web.cern.ch/gates-opening-hours>.

HR department

Warning against fraudulent telephone calls

The Swiss Mission would like to inform you that the Diplomatic Security Brigade (BSD) has recently received reports of fraudulent telephone calls from people posing as false police officers and requesting identity or bank details under the pretext of an ongoing investigation.

The Police of Geneva warns against this type of fraud and advises you to immediately stop the telephone conversation and not to communicate any personal or banking details.

The Swiss Mission also points out that the website of the National Centre for Cyber Security (NCSC) lists all cyber threats and provides practical information on the risks of fraud. Incidents can be reported directly on its website.

Report an incident:
<https://www.ncsc.admin.ch/ncsc/en/home.html>

Taxation in Switzerland

Memorandum concerning the internal tax annual certificate, the individual annual statement and income tax declaration forms in Switzerland for the reference year 2023

I – Internal tax annual certificate and individual annual statement for 2023

Your internal tax annual certificate or individual annual statement for 2023, issued by the Finance and Administrative processes Department, is available since 13 February 2024 via MyFiles (<https://myfiles.cern.ch/>) (under « Financial and Social Benefits »). The document that you have received (certificate or statement) depends on your situation at CERN in 2023. It is intended exclusively for the tax authorities.

1. If you are currently a member of the CERN personnel, you have received an e-mail containing a link to your certificate or statement, which you can print out if necessary.
2. If you are no longer a member of the CERN personnel or are unable to access your

certificate or statement as indicated above, you will find information explaining how to obtain one here: <https://cern.ch/nlegh>.

II – 2023 income tax declaration forms in Switzerland

The Admin e-guide can be found on this page: <https://cern.ch/tu5y6> to provide further general indications for completing the 2023 income tax declaration form, and to offer support in this matter.

If you have any specific questions, please contact your tax office directly.

NB: The information regarding the French income tax declaration form is usually available in April.

Contact: HR-Internal-tax@cern.ch

HR department

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

1 Mar 14:00 | Knowledge Sharing | Main Auditorium | Symposium | EN

[Herwig Schopper - a century in physics](#)

5 Mar 13:15 | Computing | CERN Science Gateway | OQI | EN

[Launch event of OQI \(Open Quantum Institute\)](#)

7 Mar 20:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN

[CERN70 public event: From particle physics to medicine](#) (more [CERN70](#) events)

8 Mar 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 9 - Intergenerational dynamics in the workplace](#)

20 Mar 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN

[Status and Prospects of the JUNO neutrino experiment](#)

Registration now open

11–19 Mar | Computing | CERN | CERN IT Storage and Data Management group | EN
[TechWeekStorageCERN24](#)

25–27 Mar | Computing | CERN | CERN openlab | EN

[2024 CERN openlab Technical Workshop](#)

15 May | Knowledge Sharing | Lausanne | Famelab | EN, FR or DE

[Famelab Switzerland local heats](#), open to 18–35-year-olds

2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN

[Mechanical & Materials Engineering for Particle Accelerators and Detectors](#)

12–25 Jun | Physics | Nakhon Pathom (TH) | Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP) | EN

[2024 Asia-Europe-Pacific School of High-Energy Physics](#)

23 Jul–1 Aug | Knowledge Sharing | European Scientific Institute (FR) | I.FAST | EN

[Accelerators for healthcare? 10-day innovation challenge](#)

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN

[Introduction to Accelerator Physics](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

CERN Library launches new newsletter to connect with CERN community

The CERN Scientific Information Service is excited to announce the launch of the CERN Library newsletter designed to strengthen ties with the CERN community.

The newsletter is available on CERN Notifications, and will deliver a curated selection of literary highlights, event announcements, and exclusive insights into the world of scientific information. Tailored to meet the reading appetites of patrons, this weekly newsletter promises to keep interested readers in the loop about the latest arrivals hitting the shelves, both in print and electronically.

This newsletter however is not just about books; it's a platform to share the richness of the library experience, from upcoming events to gems within the library collection. For example, the Meet the Author events offer the opportunity to meet notable literary figures and provide readers with a

deeper understanding of the creative minds behind the books.

You will also be informed on upcoming short training sessions organised by the Scientific Information Service, explaining how to loan books, access e-resources, submit papers in open access, and much more! The training is open to everyone: both newcomers as well as members of personnel who have been at CERN for some time can join the 30 minute training session.

Sign up for the CERN Library newsletter today and open the door to weekly literary updates, right in your inbox. Simply go to <https://cern.ch/n-wnd>, click on the 'Subscribe' button and choose the book topics you are interested in (by clicking the links available in the description). Stay connected, stay informed, and let the joy of reading unfold with each new edition.

CERN Library

TPG: disruptions buses 67 and 68 | 4 March – 20 September 2024

From 4 March until 20 September 2024, bus routes 67 and 68 will be **modified in the direction of Meyrin, Gravière / Blandonnet only**.

The “Saint-Genis, Mitterrand”, “Saint-Genis, Champ-Fusy” and “Saint-Genis, Hautains” stops **will not be served**.

The “Saint-Genis, CERN Alice” stop **will not be served** by the line 68.

The “Saint-Genis, Jean Monnet” stop on the line 68 **will be moved to the D89A**. Please go to the “Saint-Genis, Lion” stop instead.

More information on the TPG website:
<https://www.tpg.ch/en/works-and-events>.

Ombud's corner

Maintaining motivation throughout your career

Andrew* comes to my office to discuss an issue of concern. After I remind him of the working principles of the Ombud – confidentiality, informality, impartiality and independence – I invite him to explain what is worrying him. Andrew chooses to start with: “I have only six more years to work so I’m putting up with it but ...”.

The topic that I would like to explore here is not the issue that Andrew shared with the Ombud that day but the weariness and apparent lack of motivation he feels six years before retirement. Indeed, such statements from my visitors are not infrequent, and my spontaneous reaction to them is that six years, or even one year, can be very long in a job where you’ve lost meaning and motivation...

From the story that he shares with me, I see that Andrew’s **motivation was not lost overnight**. It is rarely the case. It started with difficulties with an over-controlling new supervisor, difficulties that they did not manage to overcome together. Then his request for internal mobility did not work out as quickly as he hoped. Andrew’s technical competences had been acknowledged but, as time passed, interesting projects were entrusted to younger colleagues who needed to strengthen their career trajectory, while Andrew was kept in

charge of more operational, routine tasks. A long-awaited promotion never came and that felt unfair. By the time we talked, Andrew was feeling undervalued, no longer integrated in a team and disconnected from the Laboratory’s overall mission.

There are many motivation factors at work; the key ones are:

Accountability, autonomy and empowerment at work. These are fundamental drivers of employee motivation, satisfaction and overall workplace well-being. The extent to which the Organization trusts Andrew and the freedom he is given to determine the most appropriate way to achieve clear objectives drive his motivation. The over-controlling manager and their joint inability to discuss the problem dealt the first blow to Andrew’s motivation.

Pleasant working conditions and strong connections with colleagues at work are strong drivers of motivation. A culture of equal treatment and effective conflict management are needed in the workplace to help maintain the motivation of all.

Our personal lives may at times be very demanding. This was the case when Andrew had small children and, later on, significant difficulties

with one of his teenage sons. In parallel, his workload had increased, and Andrew felt overwhelmed and not supported by his manager. His motivation dropped as he no longer could maintain a proper work–life balance. There went another chunk of motivation...

One of the strongest drivers of motivation is **continuous skills development**. The capacity to learn and develop is essential, throughout your career. When Andrew's requests for training opportunities were considered as low priority, motivation dropped a little bit more.

Fair recognition in the workplace is important and is more than just acknowledgement. Recognition includes assigning tasks that match Andrew's aptitudes and desire to learn and appreciating his investment in projects, as well as tangible rewards such as salary increments, bonuses or promotion. Pay may no longer be the main source of motivation at work. But remuneration must remain fair, in line with the skills and investment of the worker, the position held and the length of service.

Last, but not least, motivation is strongly driven by **caring and participative managers**. Managers who are available and give priority to developing their workforce and creating a cohesive and psychologically safe environment are going a long way towards keeping the members of their team motivated and invested.

Keeping a high level of motivation throughout your career is also **very much your responsibility**.

The Organization has developed a number of development modules to help you reflect on your career and on the options that you have at any point in time, such as the "Career focus", which is designed to be an energising experience.

If you feel that motivation is not as strong as it was, whatever the reasons, don't settle for weariness, boredom or frustration. It may be time for a discussion with your supervisor, your Human Resources Adviser or with the Ombud. Remember, you are the main actor in your career, but you are not alone!

There are many drivers of motivation, most of them intrinsic, and some of them are the responsibility of managers and the Organization. But it is essential, for the well-being of the workforce and the success of the Organization, to nurture and cultivate motivation throughout a career, right until the end.

* *The name and situation are fictitious.*

Laure Esteveny

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud>. More information on the role of the CERN Ombud and how to contact her can be found at <https://ombud.web.cern.ch>.